

Professor Simon Dobson FRSE CEng FBCS

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Background

Academic and professional qualifications

DPhil Computer Science, University of York UK	1993
BSc (hons) 1st class Computing Science, University of Newcastle upon Tyne UK	1989
Fellow, Royal Society of Edinburgh	2020
Fellow, British Computer Society	2008
Chartered Engineer	1996

Current position

Professor of Computer Science, University of St Andrews UK	2009 – date
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Career

University of St Andrews UK (Professor of Computer Science)	2009 – date
University College Dublin IE (College lecturer)	2004 – 2009
Aurium IE (Founder and CEO of a research-led start-up company)	2001 – 2003
Trinity College Dublin IE (Research fellow; Lecturer)	1997 – 2001, 2003 – 2004
STFC Rutherford Appleton Laboratory UK (Research fellow; Scientific officer)	1992 – 1997

Research

Current main research interests and their applications

Areas of research: Complex, self-adaptive, and sensor systems. Sensor platforms and data analytics, especially using on-device machine learning. Network-based modelling and simulation.

Areas of application: Ecological sensing, especially monitoring bird abundance using acoustics. Modelling epidemic spreading and other processes on networks.

Current and recent grant portfolio

Name	Supporter	Area	Role	Value
Cloud epyc	Oracle	Computational-science-as-a-service	PI	£30K
S4	EPSRC	Adaptive sensing	Institutional PI	£1M/£4.2M
SER	CENSIS	Environmental sensing	Institutional PI	£26K
TransM	Microsoft	Complex network epidemiology	PI	\$20K
SAPERE	EU FET	Long-lived adaptive systems	Institutional PI	€450K/€2.5M

Career totals

Principal/institutional lead investigator on grants worth over £9M. *Collaborator/co-investigator* on grants totalling a further £30M. *Commercial funding* to found and lead a start-up company, raising €650K in venture funding and growing it to a value in excess of €1.8M.

Selected recent professional activities

Academic leadership. Head of School for Computer Science, 2017 – 2021. University Research Excellence Board, 2018 – 2021. Director of Research for Computer Science, 2010 – 2017, 2022 – date. Fellowship evaluation panel, Royal Society of Edinburgh, 2021 – 2023. Member, UKCRC, 2011 – date. External examiner, Lancaster University, MSc programme, 2013 – 2017; UCD Dublin, BSc programme, 2014 – 2017. CPHC/BCS Distinguished Dissertation panel, 2010 – 2016 (chair, 2013 – 2015).

Programme and policy development. Scottish Science Advisory Council, member, 2025 – date. European Research Consortium for Informatics and Mathematics, national director 2006 – 2009 (vice-president, 2008 – 2009). Proposal reviewer, EU Framework Programmes, 2004 – 2016; EPSRC UK, 2008 – date; FCT PT, 2009. EU Framework programmes, invited expert for Situated and Autonomic Communications initiative 2003 – 2005. IBEC/ICT Ireland industry group on academic/industrial R&D, member 2003 – 2009.

Editorial activities. Associate editor, ACM Transactions on Autonomous and Adaptive Systems, 2011 – 2017. Editorial board, Journal of Network and Systems Management, 2007 – 2017; Springer Verlag Birkhäuser series on Autonomic Computing, 2008 – 2011. Guest editor, Computer Networks, 2008; Cluster Computing Journal, 2009.

Keynotes and invitations. Keynote, Conference on Design and Architectures for Signal and Image Processing (DASIP), 2018; 6th International Workshop on Managing Ubiquitous Communications and Services, 2009; 5th IEEE International Conference on Autonomic Computing, 2008; European Telecommunications Standards Institute 1st “Infinity Initiative” workshop, 2007; IFIP Autonomic Networking, 2006. Invited lecture series, Swiss doctoral winter school, 2012.

Conference and workshop organisation. IEEE International Conference on Autonomic and Self-Organising Systems (ACSOS), Local chair 2006; workshops chair 2007; programme co-chair 2008; general co-chair 2009; member of steering committee 2007 – date.

Selected recent publications

Berné Nortier, Simon Dobson, and Federico Battiston. Higher-order shortest paths in hypergraphs. *Physical Review E*, 112, 2025

Peter Mann, Lei Fang, and Simon Dobson. Mixing patterns in graphs with higher-order structure: the role of inter-subgraph correlations. *Journal of Physics: Complexity*, 6(4), 2025

Tianyuan Zheng, John Mitchell, and Simon Dobson. Revisiting the application of machine learning approaches in predicting aqueous solubility. *ACS Omega*, 9(32), 2024

Peter Mann, V. Anne Smith, John Mitchell, and Simon Dobson. N-strain epidemic model using bond percolation. *Physical Review E*, 106, 2022

Peter Mann, V. Anne Smith, John Mitchell, and Simon Dobson. Two-pathogen model with competition on clustered networks. *Physical Review E*, 103(6), June 2021

Peter Mann, V. Anne Smith, John Mitchell, and Simon Dobson. Co-operative co-infection dynamics on clustered networks. *Physical Review E*, 103(4), April 2021

Peter Mann, V. Anne Smith, John Mitchell, and Simon Dobson. Random graphs with arbitrary clustering and their applications. *Physical Review E*, 103(1), January 2021

Simon Dobson. *Epidemic modelling – Some notes, maths, and code*. Independent Publishing Network, 2020

Simon Dobson, Matteo Golfarelli, Simone Graziani, and Stefano Rizzi. A reference architecture and model for sensor data warehousing. *IEEE Sensors Journal*, 18, 2018

Danilo Pianini, Simon Dobson, and Mirki Viroli. Self-stabilising target counting in wireless sensor networks using Euler integration. In *Proceedings of the Eleventh IEEE International Conference on Self-Adaptive and Self-Organizing Systems (SASO’17)*, pages 11–20, September 2017

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